

Springer Texts in Statistics

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An Introduction to Statistical Learning

with Applications in R

 Springer

ANALYTICS

Week 1

Instructor:

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*Examples Inspired By
CAS, SOA, Coaching
Actuaries ©*

Source Text

- The concepts are the hardest part of SRM.
- There's so many of them and they come fast.
- You **have to** read the source text to get the details for understanding.

About Notes

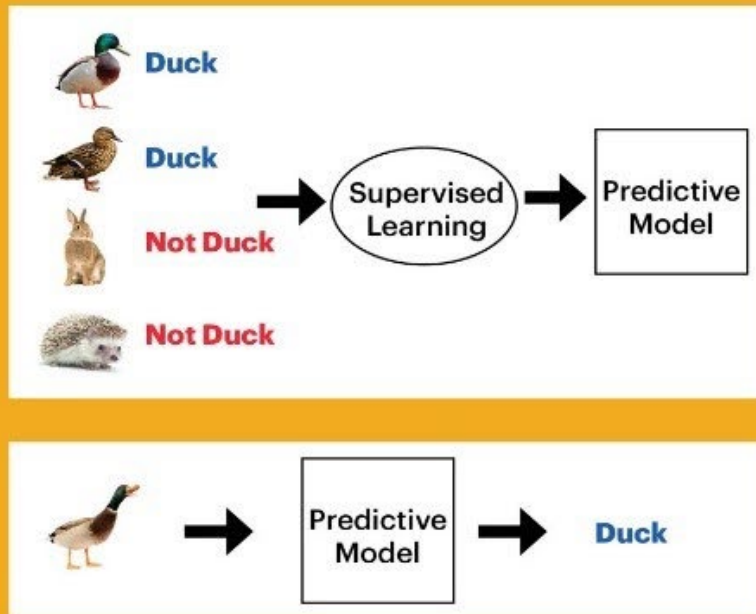
- It is very important to have the notes loaded into a tablet or computer via an app like OneNote to take notes from or print them out, since you will not be able to write everything down from scratch and think enough about the topic.

Statistical Learning

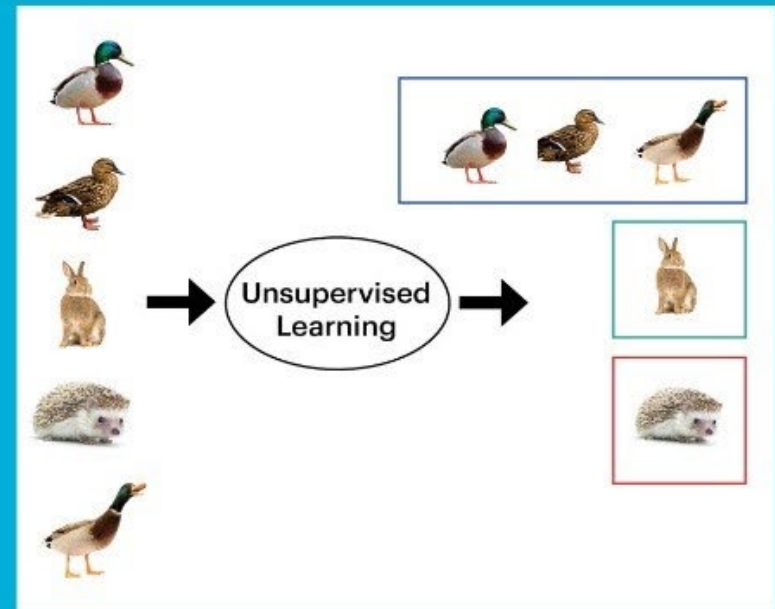
- **Statistical/machine learning:** Tools to understand data
- **Supervised learning:** _____
- **Unsupervised learning:** All data are inputs, and goal is to understand relationships and structure of data

Types of Learning

Supervised Learning (Classification Algorithm)



Unsupervised Learning (Clustering Algorithm)



Supervised Learning

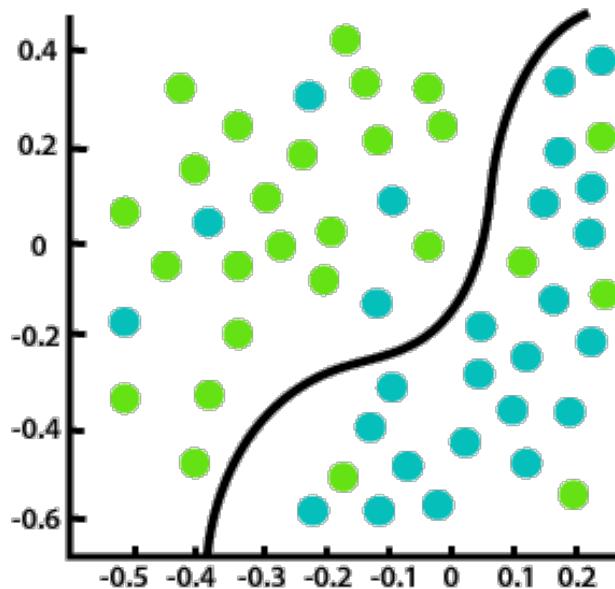
- In supervised learning, the **response variable** (Y) is the _____
- **Explanatory variables** are used as _____

Supervised Learning 2

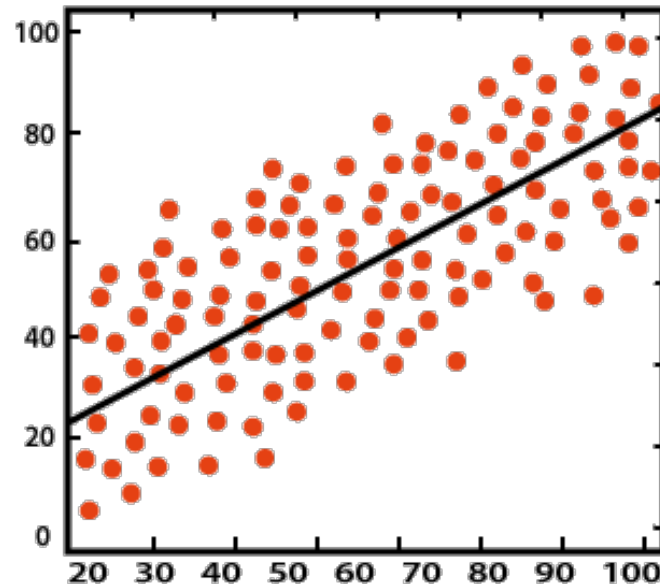
- Goal: Use explanatory variables (inputs) from the dataset X to predict the response variable Y .
- **Regression problem:** Y is _____
- **Classification problem:** Y is _____

Regression v. Classification

- <https://www.javatpoint.com/regression-vs-classification-in-machine-learning>



Classification



Regression

Quantitative Variables

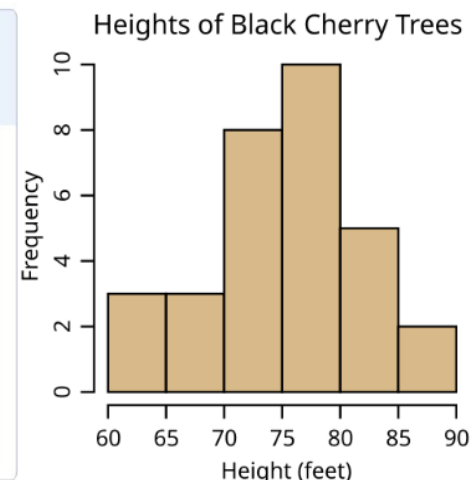
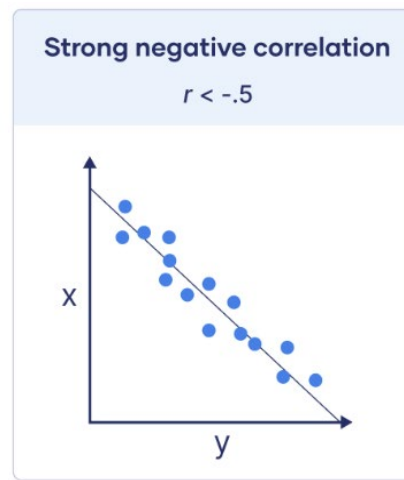
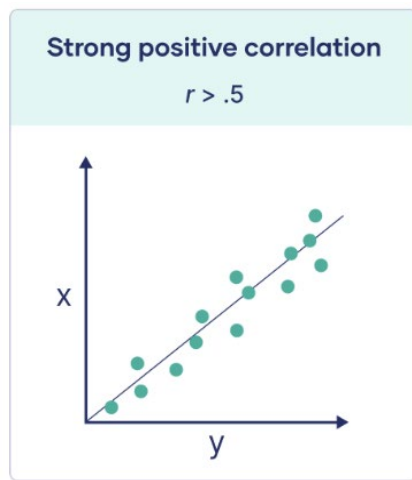
- **Continuous:** Any real value in the domain
(Regression most appropriate)
- **Count:** Assumes non-negative integer values
(special regression techniques appropriate)

Qualitative Variables

- **Categorical/Qualitative:** Assumes a small # of categories (Classification most appropriate)
- **Nominal (Categorical):** _____
- **Ordinal (Categorical):** _____

Exploratory Data Analysis

- Before modeling, explore your data!
- Histograms: Show each variable's distribution
- Summary Statistics: Understand data structure
- Pearson Correlation r :



Terminology

- X Input variables = predictors, independent variables, inputs, features
- Y Output variable = response variable, dependent variable, target
- $Y = f(X_1, X_2, \dots, X_p) + \epsilon$
 - ϵ : random error term / idiosyncratic part, assumed mean 0, variability is irreducible
 - f : systematic part (X provides info about Y), does “something” to X

Classification

- In classification problems, $f(X_1, X_2, \dots, X_p)$ produces the most likely category/class, commonly using the _____

Errors

- $E(Y - \hat{Y})^2 = [f(x) - \hat{f}(X)]^2 + Var(\epsilon)$
- **Reducible error:** $[f(x) - \hat{f}(X)]^2$
- **Irreducible error:**

Prediction vs. Inference

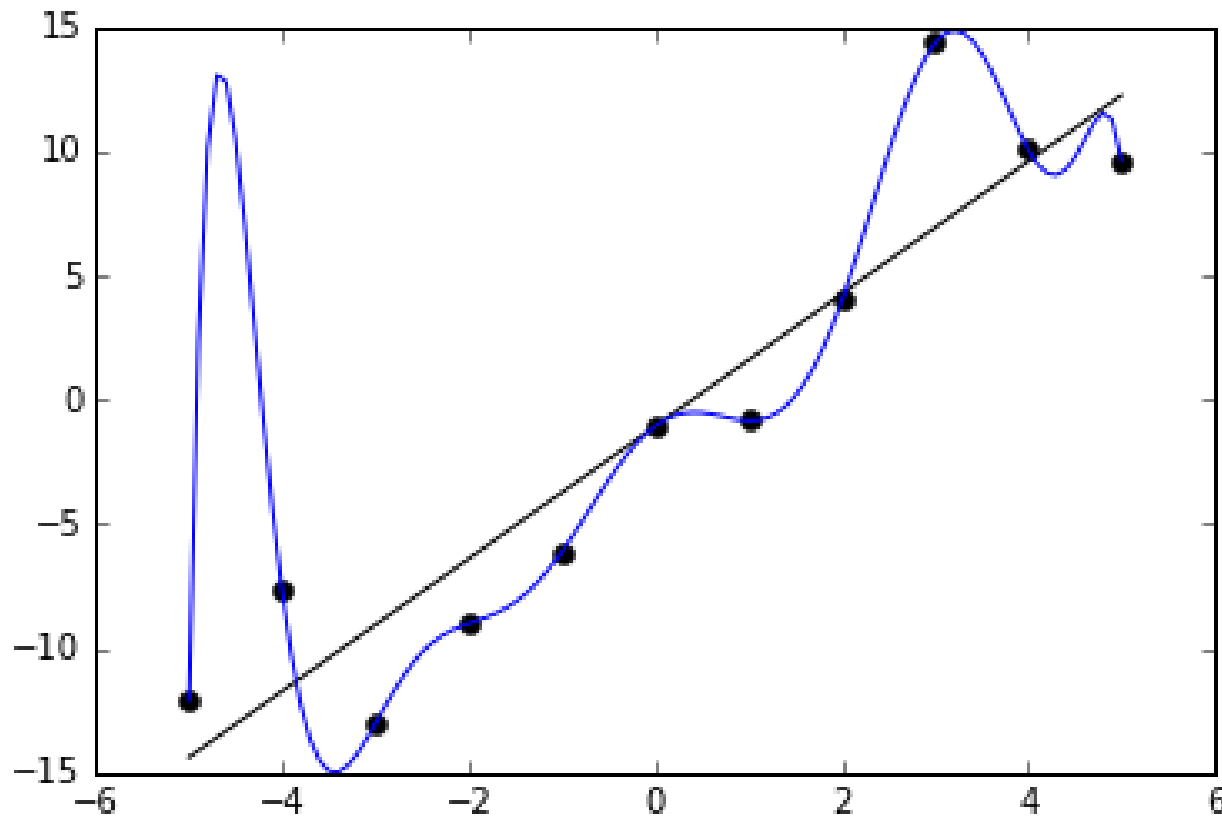
- **Prediction:** _____
- **Inference:** Determine how Y changes as the inputs change.
- **Inference:** Identify the most important predictors.
- **Inference:** Find the relationship between Y and each predictor.

Parametric Models

- Functional form of f is pre-set, so _____
- Con: May not match true f , but greatly simplifies estimation process
- More parameters used will add flexibility, but increase risk of _____

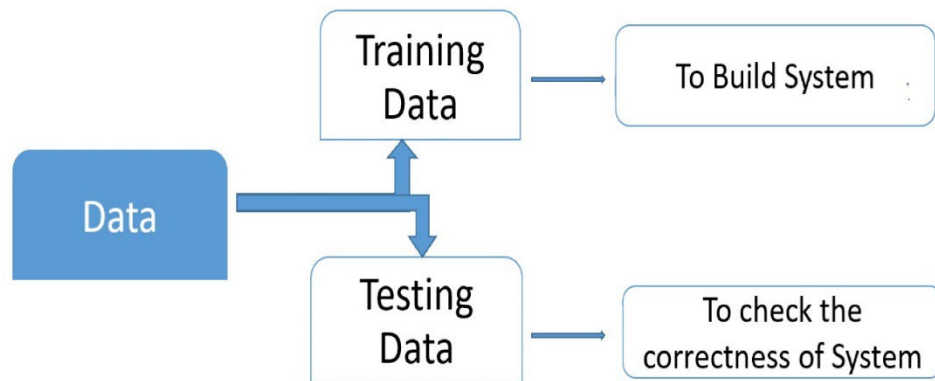
Overfitting

- The polynomial model is fitting to noise.



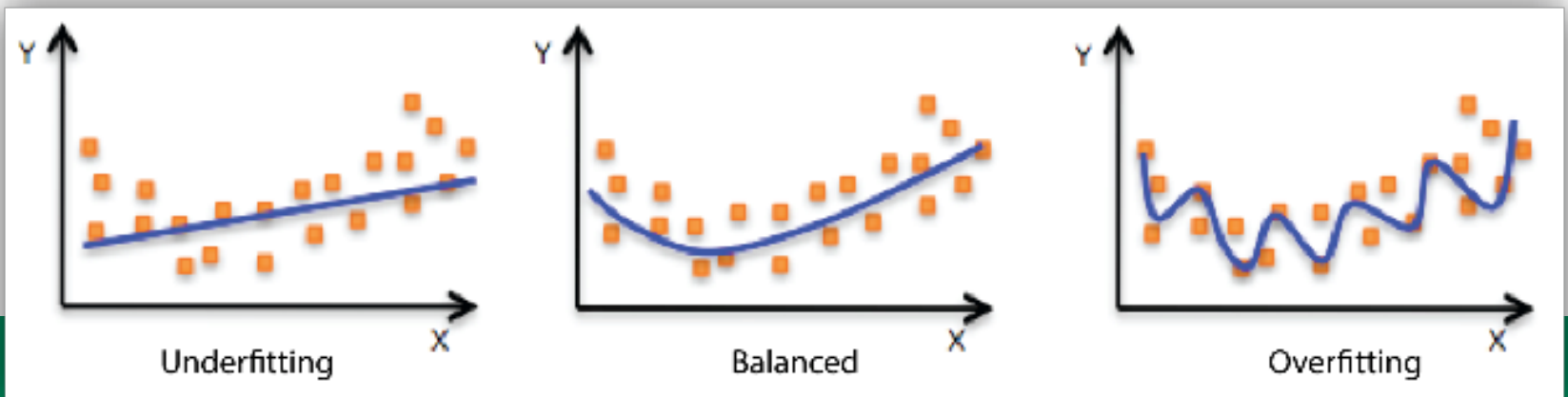
Training vs Testing

- Use training data to _____
- Use testing data to _____



Over vs. Under-fitting

- Underfitting = Model too simple =
- Overfitting = Model too complex =



Non-Parametric Models

- Determine a function $\hat{f}$ that models _____
- Con: Much larger # of observations needed to create an accurate function vs. parametric
- Lower smoothness will increase risk of overfitting.

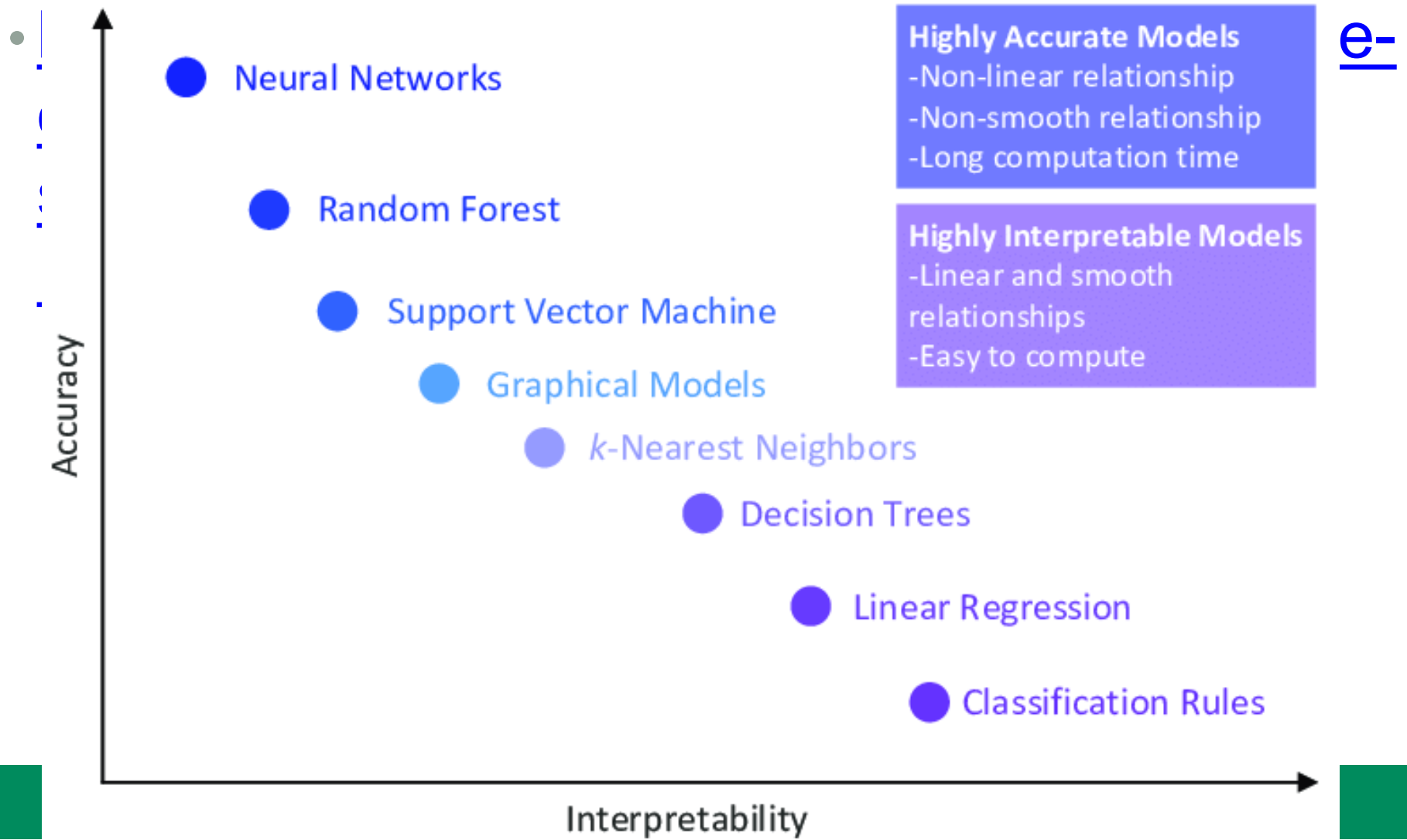
Flexibility

- Models that are more **flexible** have more free parameters to estimate and might be able to fit closer to the data.
- Flexible models might improve _____

Accuracy vs. Interpretability

- If prediction accuracy is the goal, a more _____
- If inference and study of relationships is the goal, a more _____
- Higher flexibility model → More “black-box” typically

Accuracy Interpretability



Model Principles

- A key goal is to find the right # of fitted parameters such that the model not under-fit nor over-fit.
- The principle of parsimony prefers a simpler good enough model vs. a black-box hard to understand model that _____

Model Types Summary

Machine / Statistical Learning

Supervised (Y)

Unsupervised

Regression (#)

Classification

Parametric (f)

Non-Parametric

- Key Considerations:
- Is prediction or inference more important?
- Is flexibility or interpretability more important?

Models Covered

- **Bold** = On MAS-I, *Italics* = On Exam PA

Supervised Parametric	Supervised Non-Parametric
Simple Linear Regression	K-Nearest Neighbors
Multiple Linear Regression	<i>Regression Trees</i>
<i>Generalized Linear Models</i>	<i>Classification Trees</i>
<i>Logistic/Probit Regression</i>	<i>Bagging, Random Forests</i>
<i>Poisson Count Regression</i>	
Time Series Trend Models	Unsupervised Models
Time Series Autoregressive	<i>Principal Components</i>
Ridge/LASSO Regression, Partial Least Squares	<i>Clustering</i>

Poll Concept Check #1

- Which statements are true?
- I. In $Y = f(X) + \varepsilon$, $f(X)$ represents the signal component.
- II. More data is required to fit a parametric model vs. a non-parametric one.
- III. Finding the largest risk factor for student drop-out is an example of prediction.
- IV. A more flexible model may not result in better prediction accuracy.
- V. Classification problems deal with qualitative predictors.
- VI. Grouping viewers by movies they like is an example of unsupervised learning.
- VII. Categorical variables take on one of a finite number of options.

Measures of Fit

- After fitting a model, several tools are available to numerically describe how it fits the data.
- The most popular metric for a quantitative Y is mean squared error (**MSE**).
- The goal of any model is to minimize the **test data** MSE (withheld data from training), as opposed to the **training data** MSE, which _____

MSE

- Mean Squared Error $MSE = E \left[(Y - \hat{Y})^2 \right] =$
- n : # of observations in dataset (typically training or test)
- y_i : Response for observation i
- $\hat{y}_i$: Prediction for observation i

MSE 2

- Minimizing MSE on the training dataset will lead to overfitting.
- Overfitting occurs when the model follows training data too closely instead of _____
- Thus, the goal of statistical learning is to minimize MSE of the test data, using a model built on the _____

MSE 3

- <https://medium.com/greyatom/what-is-underfitting-and-overfitting-in-machine-learning-and-how-to-deal-with-it-6803a989c76>



Test vs. Training MSE

- Test *MSE* often follows a U-shaped curve as flexibility (# of estimated parameters) increases.
- Training *MSE* decreases monotonically as model flexibility increases.

Classification Metrics

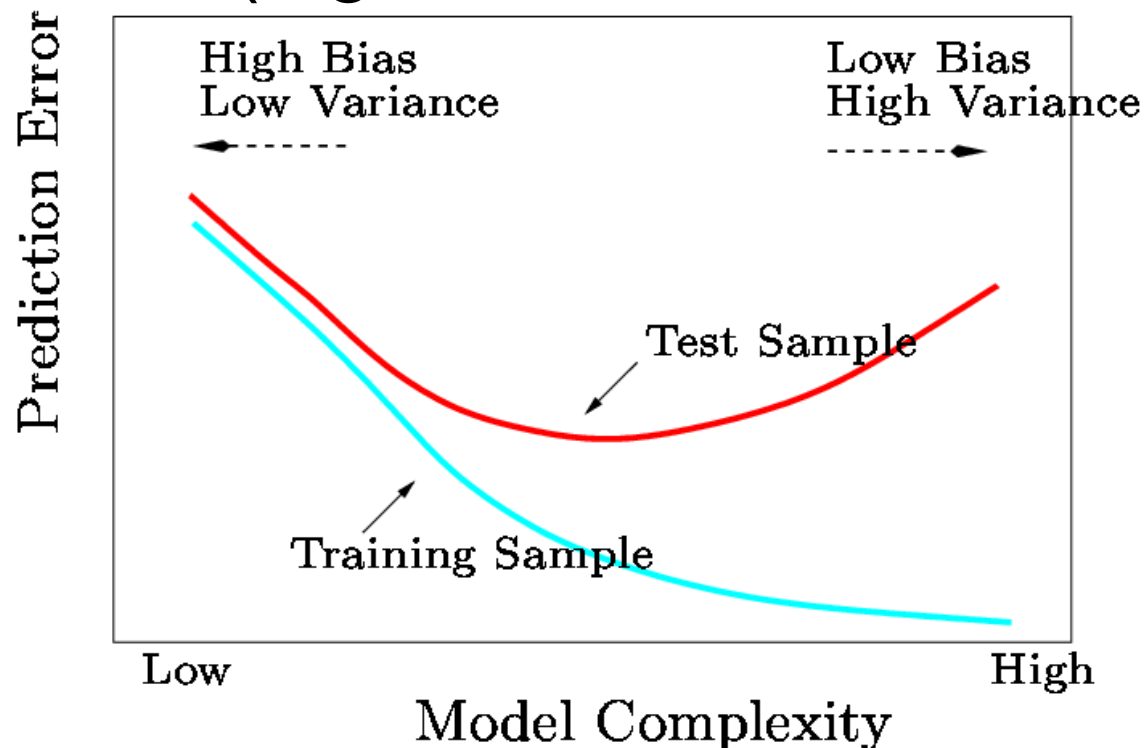
- For categorical Y , different approaches for quantifying the accuracy of $\hat{f}$ are needed.
- Error Rate: $\frac{1}{n} \sum_{i=1}^n I(y_i \neq \hat{y}_i)$
 - I : **Indicator variable** that equals 1 (incorrect prediction) or 0 (correct)
- Goal: Minimize the test error rate

Bayes Classifier

- The test error rate is minimized by the Bayes classifier.
- Assign observation to the class with the
- Bayes error rate: $1 - E \left[\max_j \Pr(Y = j|X) \right]$

Bias-Variance Trade-Off *

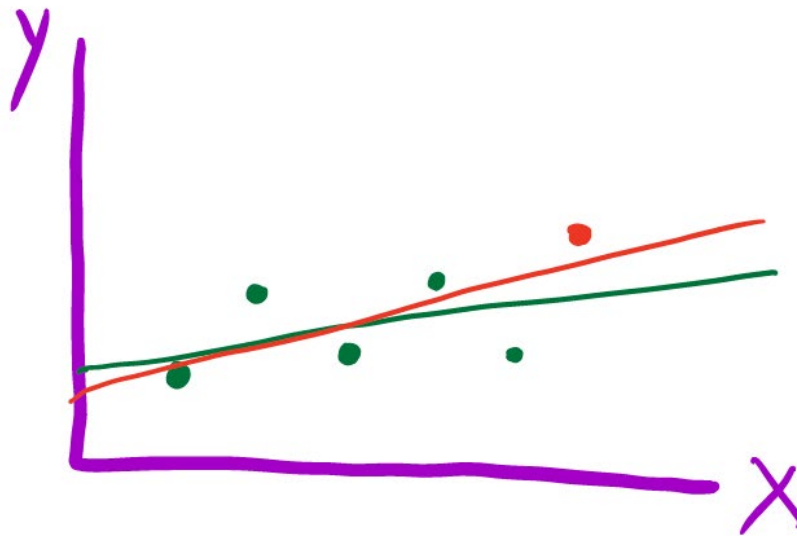
- Training: More is always better
- Test: More eventually = overfitting
- No free lunch (higher bias = lower variance)



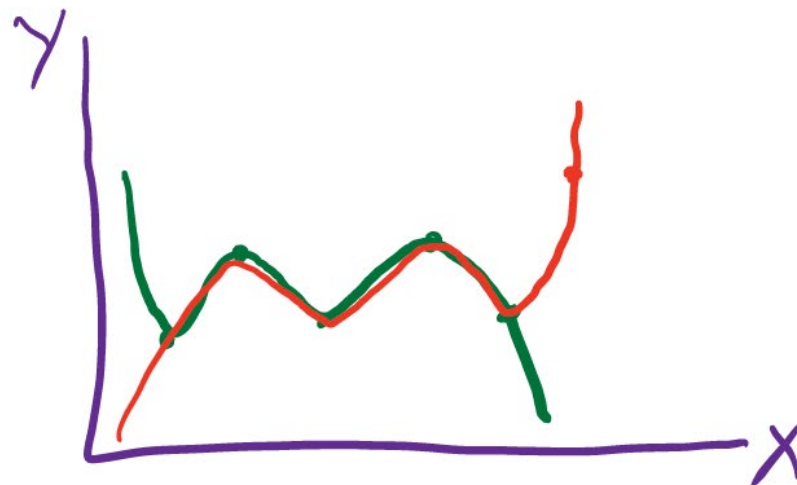
Bias-Variance Trade-Off

- $MSE = Var(\hat{f}(X)) + \text{Bias}(\hat{f}(X))^2 + Var(\epsilon)$
 - $Var(\hat{f}(X))$: Model variance
 - $\text{Bias}(\hat{f}(X)) = E[\hat{f}(X)] - f(X)$
 - $Var(\epsilon)$: Irreducible error
- Variance: Change in model from
- Bias: Error arising from the model used deviating from true model

Model Variance



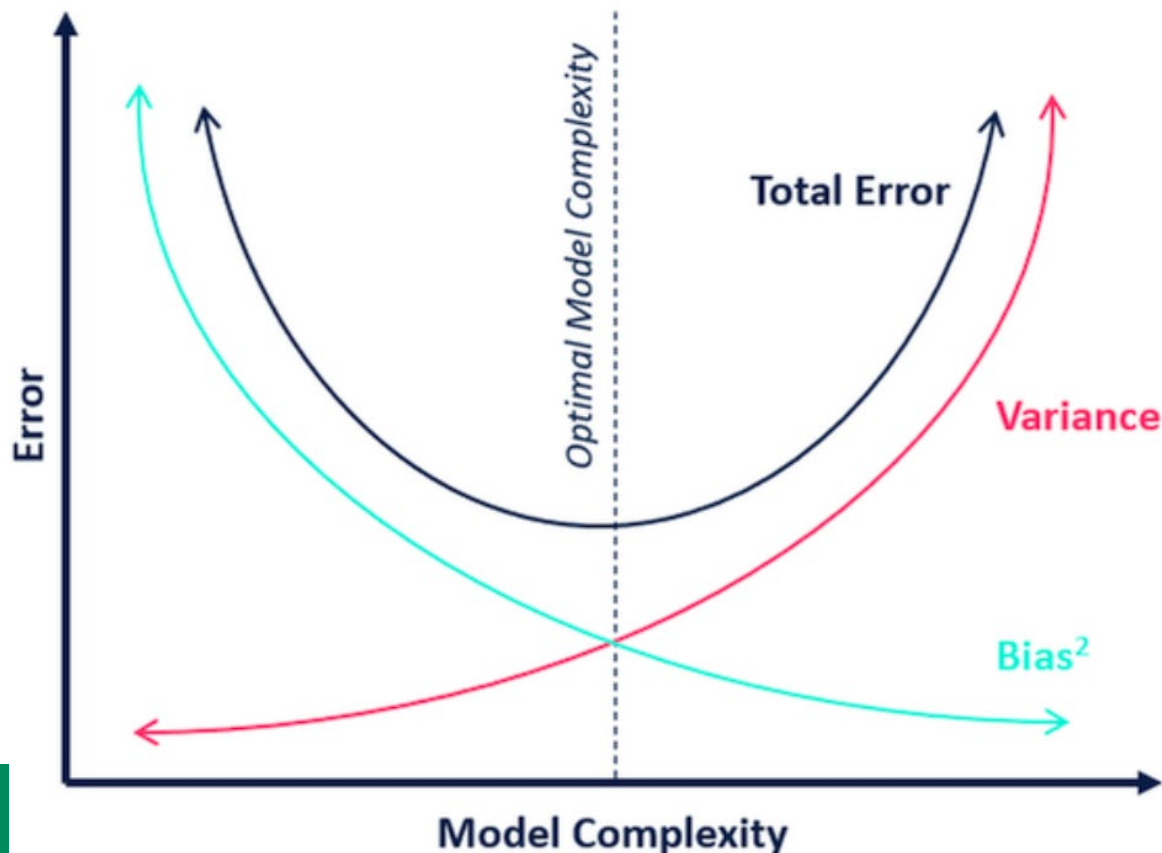
low model
variance



high model
variance

Bias-Variance Trade-Off 2

- <https://ai-pool.com/a/s/bias-variance-tradeoff-in-machine-learning>



Poll Concept Check #2

- For each item, as flexibility increases, does the item monotonically increase, monotonically decrease, or neither?
- I. Test MSE
- II. Training MSE
- III. Bias Squared
- IV. Variance
- V. Irreducible error

Degrees of Freedom

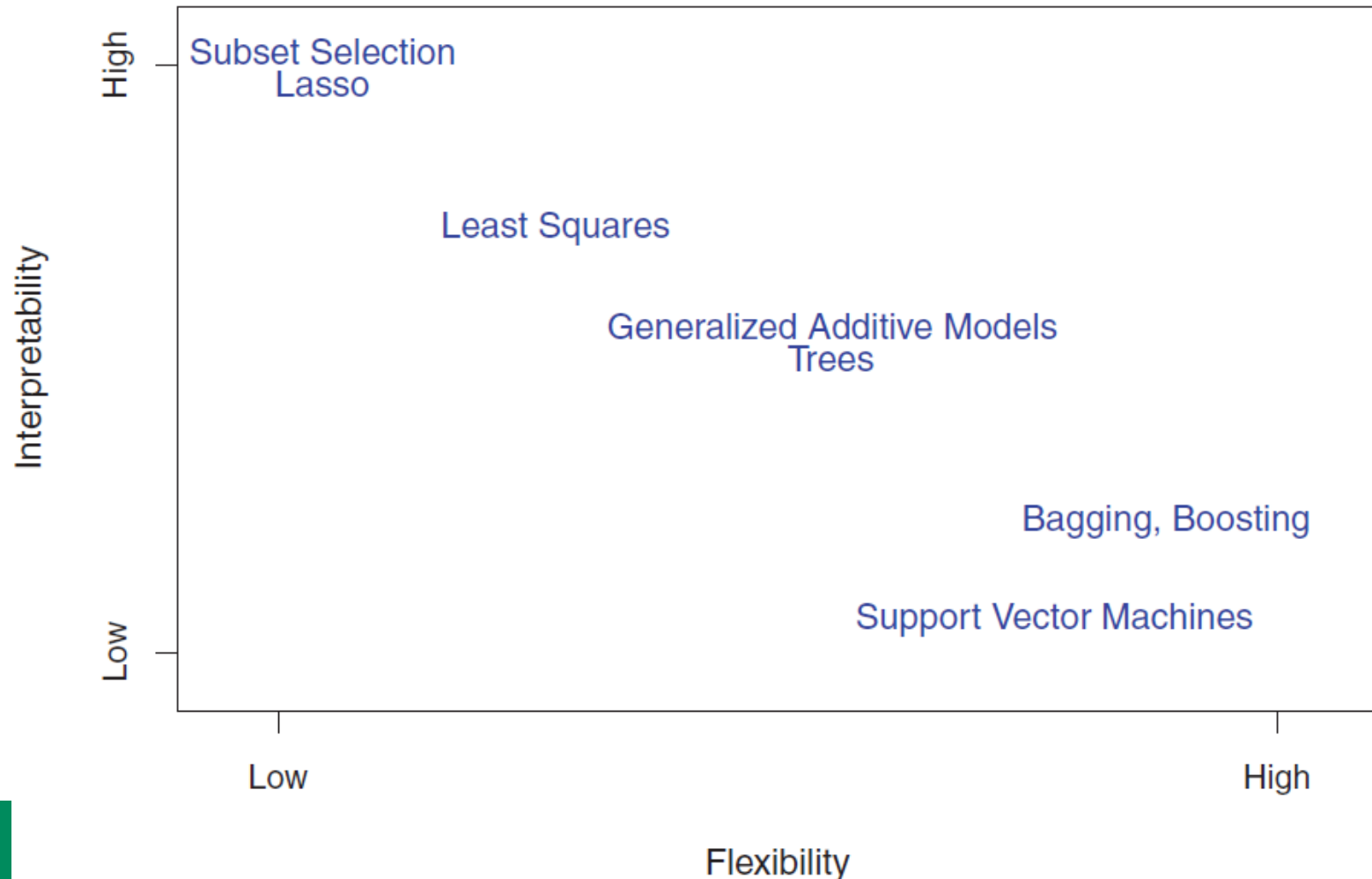
- Degrees of freedom helps quantify model flexibility
- Fewer df used (more df remaining) →

Bias-Variance Summary

- As model flexibility increases, variance will increase and bias will decrease.
- Low flexibility = _____
- High flexibility = _____
- Expected Test $MSE \geq Var(\epsilon)$ as irreducible error, natural variability is never eliminated

Common Models

- © ISL



Poll Concept Check #3

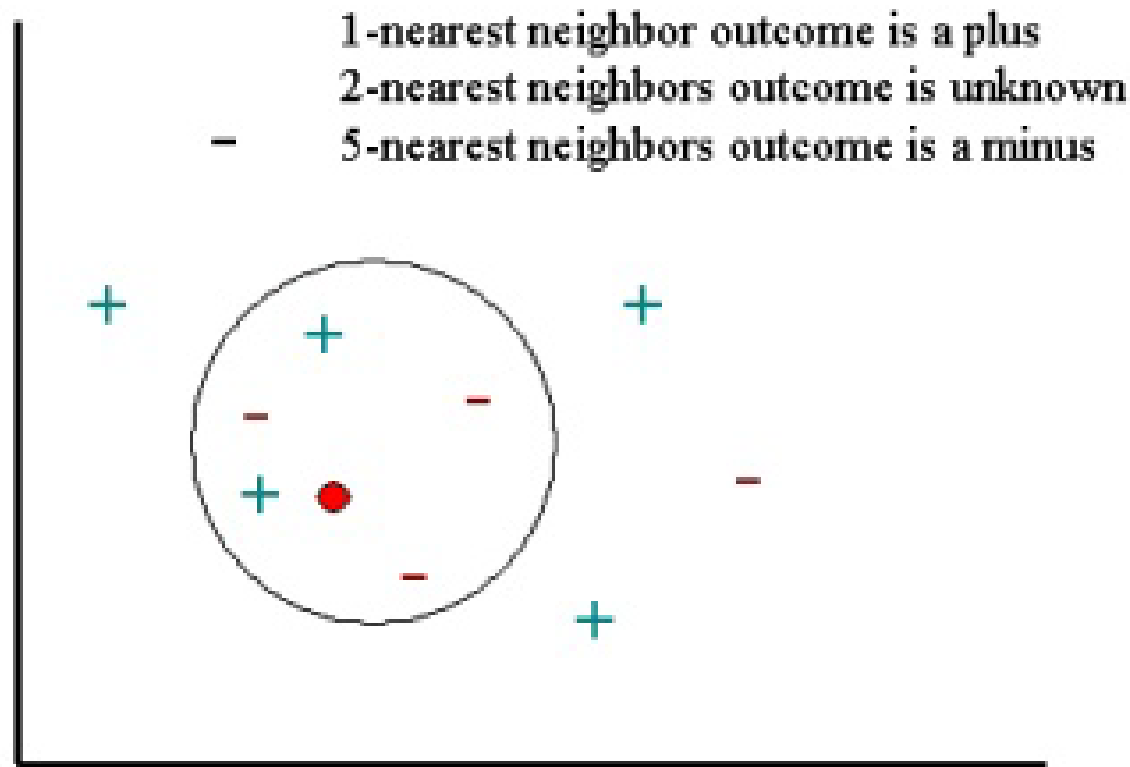
- ISL Chapter 2 Q1
- For each part, indicate whether we would generally expect the performance of a flexible statistical learning method to be better or worse than an inflexible method.
- A. Sample size n is extremely large, and # of predictors p is small
- B. The # of predictors p is extremely large, and the # of observations n is small
- C. The relationship between the predictors and response is highly non-linear
- D. Variance of the error terms $Var(\varepsilon)$ is extremely high

K-Nearest Neighbors

- KNN is a very simple, elegant technique that says to categorize an observation x_0 into one of several classes, look at the “nearest” K points and _____
- It's a _____

K-Nearest Neighbors 2

- Categorize the unknown based on known similar points



K-Nearest Neighbors 3

- For each test observation x_0 and a set positive integer K , the KNN classifier, identifies the K points in the training data (N_0) closest to x_0 .
- Then, it determines the conditional probability x_0 is in class j as the proportion of points in N_0 whose response values equal j .
- Finally, apply Bayes Theorem to assign x_0 to the class with highest probability.
- K is inversely related to flexibility while $\frac{1}{K}$ is directly related to flexibility.

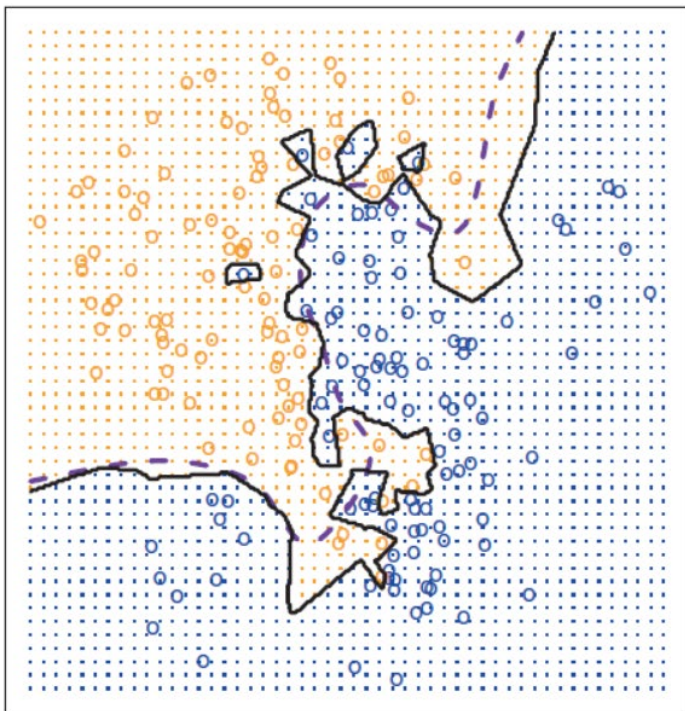
Distance

- If unspecified, “distance” and “closeness” in multi-dimensions means Euclidean distance.
- $d(\mathbf{p}, \mathbf{q}) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$
 - $\mathbf{p}, \mathbf{q}$: coordinate vector of two respective points
- In 2 dimensions, this is the distance between 2 points (p_1, p_2) and (q_1, q_2) formula:
- $d(\mathbf{p}, \mathbf{q}) =$

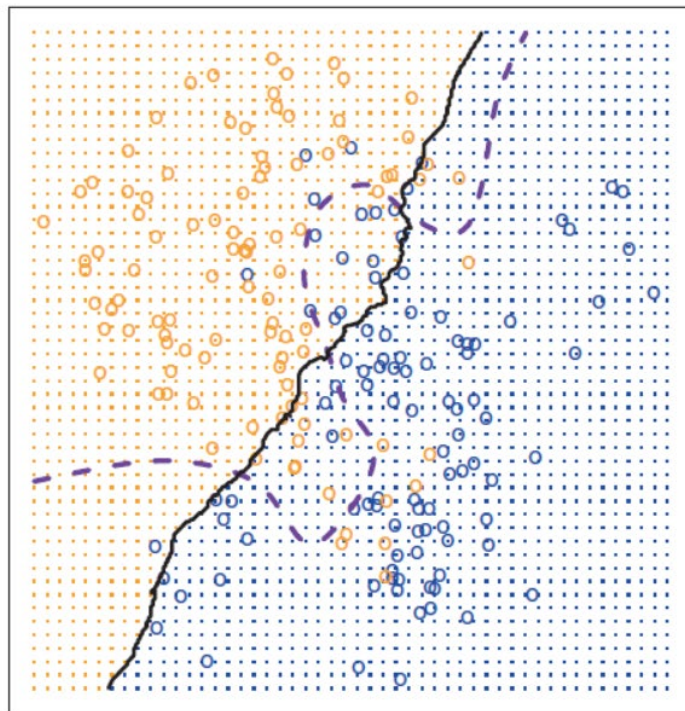
K-Nearest Neighbors 4

- Small k : jagged, more flexible, higher variance
- Large k : smooth, higher bias, lower variance

KNN: K=1



KNN: K=100



KNN Regression

- KNN regression works on regression problems

- The prediction $\hat{y}$ is the _____

HW Note

- This week's CA HW is primarily conceptual, and you are expected to read the Coaching Review Guide on Canvas Files.
- **There WILL be many Qs on topics (trees, ridge, lasso, etc.) we didn't cover. You won't be responsible until we cover the topic.**

HW Note Factoids

- Decision trees can classify observations into groups using labeled data, are supervised, and are generally non-parametric.
- Ridge/Lasso are typically more interpretable than black-box methods.
- KNN/boosting/trees can be used for both regression and classification contexts.

1. Classifier Illustration

- The Bayes classifier is used to determine if Team 1 wins a soccer game match based on two predictors. X_1 represents Team 1 strength, and X_2 Team 2 strength. The table shows Team 1 win probability for every equally likely combination.
- Calculate the Bayes error rate.

	$X_2 = 1$	2	3	4
$X_1 = 1$	0.46	0.4	0.35	0.2
2	0.55	0.48	0.45	0.3
3	0.75	0.65	0.52	0.4
4	0.8	0.7	0.6	0.54

1. Classifier Illustration 2

- Bayes error rate: $1 - E \left[\max_j \Pr(Y = j|X) \right]$

	$X_2 = 1$	2	3	4
$X_1 = 1$				
2				
3				
4				

1. Classifier Illustration 3

- $E \left[\max_j \Pr(Y = j|X) \right] = \frac{\text{Sum of Prob. Right}}{\text{Max Correct}}$

2. KNN Illustration

- ISL Ch. 2 Q7
- The table below provides a training dataset with 6 observations, 3 predictors, and 1 qualitative response variable.
- Make a prediction for Y when $X_1 = X_2 = X_3 = 0$ using K -nearest neighbors, with $K = 3$.

Obs.	X_1	X_2	X_3	Y
1	0	3	0	Red
2	2	0	0	Red
3	0	1	3	Red
4	0	1	2	Green
5	-1	0	1	Green
6	1	1	1	Red

2. KNN Illustration 2

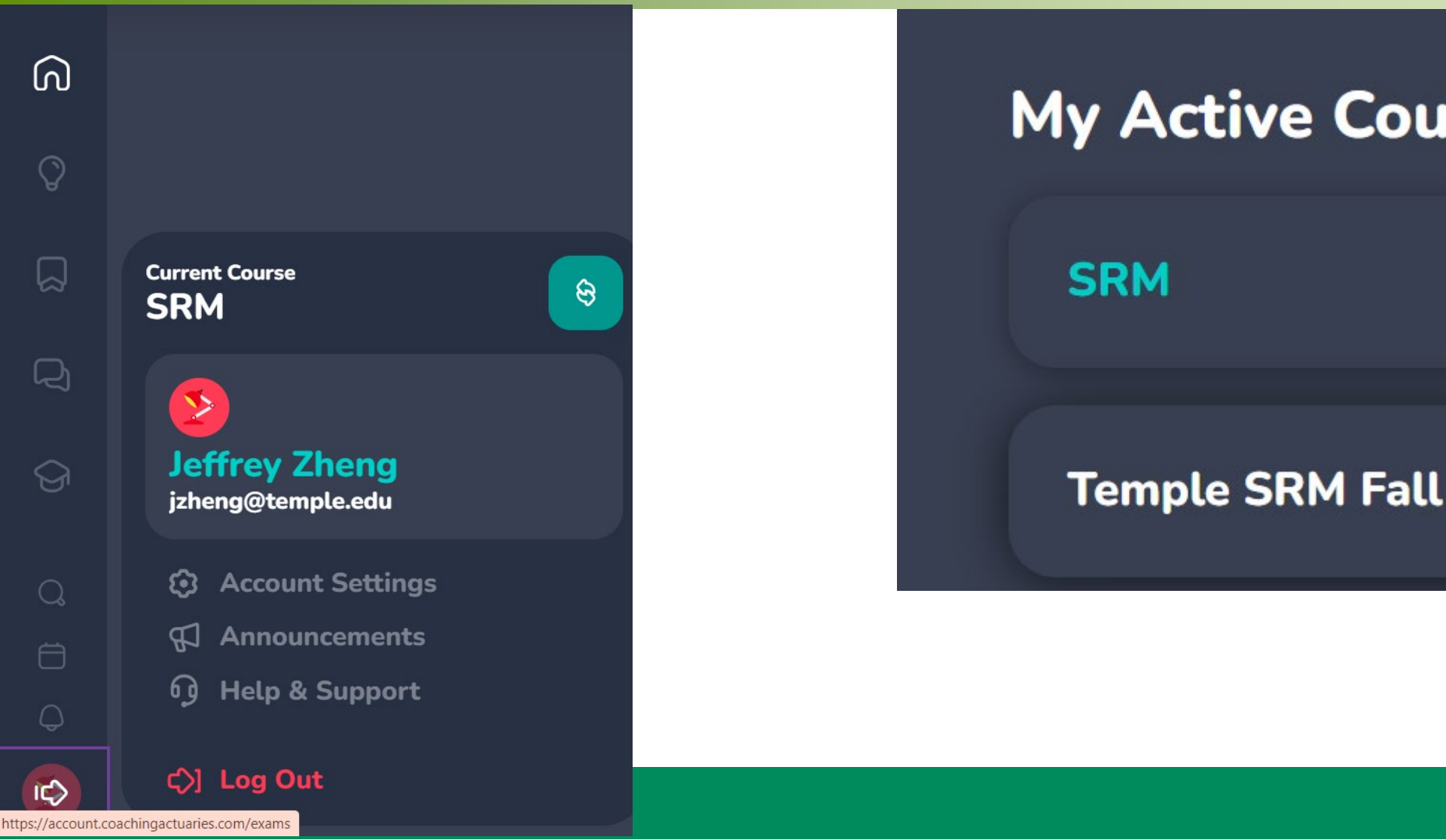
- $d(\mathbf{p}, \mathbf{q}) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$
- $\Pr(Y = j | X = x_0) = \frac{1}{K} \sum_{i \in N_0} I(y_i = j)$

3. KNN Regression

- Make a prediction for Y when $X_1 = X_2 = X_3 = 0$ using K -nearest neighbors, with $K = 3$.

Obs.	X_1	X_2	X_3	Y
1	0	3	0	0
2	2	0	0	1
3	0	1	3	3
4	0	1	2	5
5	-1	0	1	6
6	1	1	1	2

Finding Coaching HW



Summary Check

- *Can you describe in your own words:*
- Key Terms: Response, Explanatory, Count, Continuous, Nominal, Ordinal
- Regression vs. Classification
- Parametric vs. Non-Parametric
- Flexibility vs. Interpretability
- Prediction vs. Inference
- Training vs. Test Data, MSE shapes
- Bias-Variance trade-off
- Under- vs. Over-fitting
- Categorize popular methods
- How KNN works